

Statement of Verification

BREG EN EPD No.: 000255

Issue 04

This is to verify that the

Environmental Product Declaration

provided by:

Altro Ltd



is in accordance with the requirements of:

EN 15804:2012+A1:2013

and

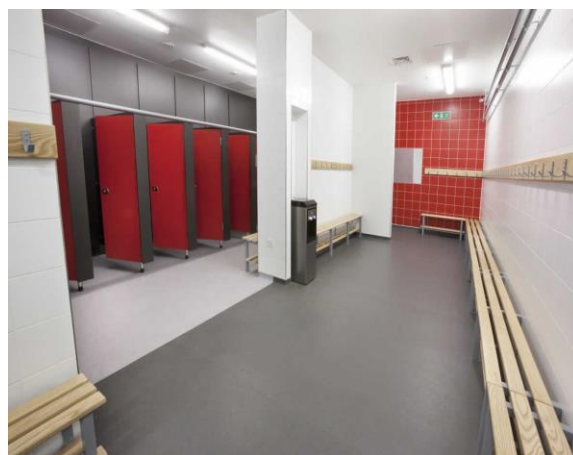
BRE Global Scheme Document SD207

This declaration is for:

Altro Standard Safety Floor Products, 2 to 2.5mm

Company Address

Works Road
Letchworth Garden City
Hertfordshire
SG6 1NW
United Kingdom



DPatel

Deep Patel
Operator

23 August 2024
Date of this Issue

09 March 2019
Date of First Issue

22 December 2024
Expiry Date



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Environmental Product Declaration

EPD Number: 000255

General Information

EPD Programme Operator	Applicable Product Category Rules
BRE Global Watford, Herts WD25 9XX United Kingdom	BRE Environmental Profiles 2013 Product Category Rules for Type III environmental product declaration of construction products to EN 15804:2012+A1:2013
Commissioner of LCA study	LCA consultant/Tool
Altro Ltd Works Road Letchworth Garden City Hertfordshire SG6 1NW United Kingdom	Fei Zhang / BRE LINA v2.0
Declared/Functional Unit	Applicability/Coverage
1m ² of PVC flooring	Manufacturer specific product range
EPD Type	Background database
Cradle to Gate	ecoinvent v3.2
Demonstration of Verification	
CEN standard EN 15804 serves as the core PCR ^a	
Independent verification of the declaration and data according to EN ISO 14025:2010 ☐ Internal ☒ External	
(Where appropriate ^b) Third party verifier: Nigel Jones	
a: Product category rules b: Optional for business-to-business communication; mandatory for business-to-consumer communication (see EN ISO 14025:2010, 9.4)	
Comparability	
Environmental product declarations from different programmes may not be comparable if not compliant with EN 15804:2012+A1:2013. Comparability is further dependent on the specific product category rules, system boundaries and allocations, and background data sources. See Clause 5.3 of EN 15804:2012+A1:2013 for further guidance	

Information modules covered

Product			Construction		Use stage							End-of-life				Benefits and loads beyond the system boundary
A1	A2	A3	A4	A5	Related to the building fabric					Related to the building		C1	C2	C3	C4	D
Raw materials supply	Transport	Manufacturing	Transport to site	Construction – Installation	Use	Maintenance	Repair	Replacement	Refurbishment	Operational energy use	Operational water use	Deconstruction demolition	Transport	Waste processing	Disposal	Reuse, Recovery and/or Recycling potential
☒	☒	☒	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

Note: Ticks indicate the Information Modules declared.

Manufacturing site

Altro Ltd
Works Road
Letchworth Garden City
Hertfordshire
SG6 1NW
United Kingdom

Construction Product:

Product Description

This product range covers 2.0 mm to 2.5 mm thick sheet PVC based Standard Safety Flooring without PUR Lacquer, to EN 13845. This product range representative EPD covers the products **Altro ContraX**, **Altro Classic 25**, **Altro Marine 20**, **Altro Impressionist II**.

Technical Information

The below table covers the basic technical properties of the four products within the 2.0 mm to 2.5 mm thick sheet PVC based Standard Safety Flooring without PUR Lacquer product range. For further properties, please see the specific product's page on Altro's website <https://www.altro.co.uk>.

Property	Altro ContraX	Altro Classic 25	Altro Marine 20	Altro Impressionist II
Thickness (EN 428)	2 mm	2.5 mm	2.0 mm	2.0 mm
Mass per area (EN 430)	2.5 kg/m ²	3.2 kg/m ²	2.5 kg/m ²	2.6 kg/m ²
Slip resistance (TRRL) (EN 13845) (EN 13893) (DIN 51130 / DIN 51097 (Altro Marine 20 Only))	≥ 36 ESf DS R10	≥ 45 ESf DS R11	ESb DS R10 / Class C	≥ 36 ESf DS R10
Fire performance (EN 13501-1, EN ISO 9239-1, EN ISO 11925-2) CAN/ULC	Class Bfl-s, ≥8kW/m ² , pass S102.2 (Tested)			

ASTM E648 ASTM E662	Class 1 <450
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Main Product Contents

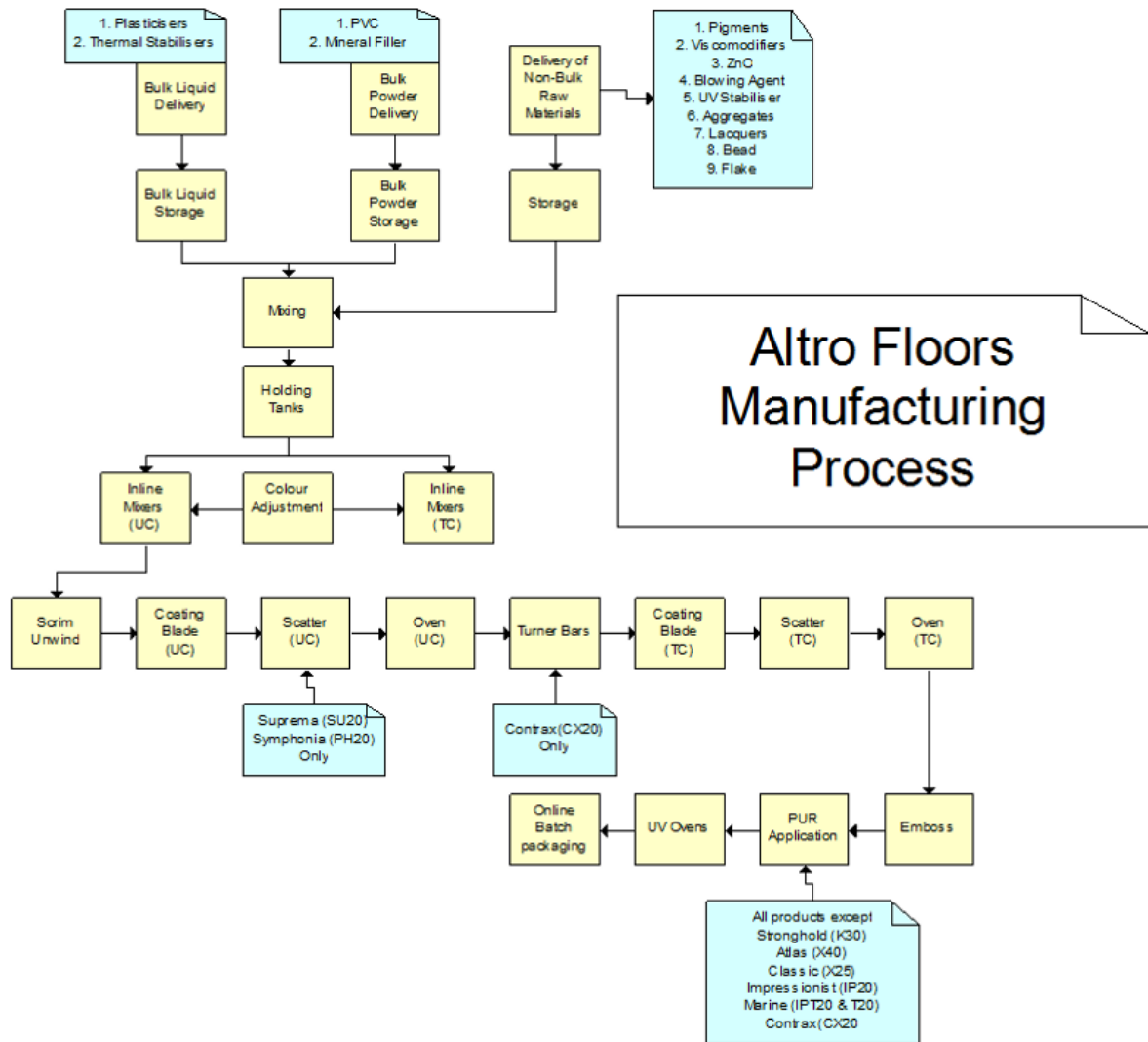
The raw material composition of the product range covered by this EPD is given below.

Material/Chemical Input	Mass (%)
Plastisol	90 - 94
Scatter	4 - 8
Scrim	2

Manufacturing Process

Bulk liquids, powders, performance additives and some aggregates are mixed together into a plastisol and placed in a holding tank. The plastisol is then pigmented and passed into inline mixers. The plastisol is then coated onto a scrim and aggregates are scattered onto the surface to aid slip resistance and durability. The product is then cured in an oven, cut into rolls and packaged for dispatch.

Process flow diagram



Life Cycle Assessment Calculation Rules

Declared / Functional unit description

1m² of Altro Classic 25 2.5mm thick Altro standard safety flooring without PUR lacquer. The declared unit represents the Altro ContraX, Altro Classic 25, Altro Marine 20, and Altro Impressionist II products for 2.0 – 2.5mm thicknesses.

System boundary

This is a cradle-to-gate EPD, reporting all production life cycle stages (modules A1 to A3) in accordance with EN 15804:2012+A1:2013.

Data sources, quality and allocation

The supporting LCA study was carried out using BRE LINA v2.0 using manufacturer specific data provided by Altro for the production period of the 12 months of 2017 at the Letchworth site.

The Letchworth site produces other PVC products in addition to the 2.0 – 2.5mm standard safety flooring without PUR lacquer product range, so allocation was applied to site wide values for packaging, energy, water, non-production waste, and wastewater, on a m² of production basis. Production waste was allocated on a percentage mass of production basis. No allocation of raw material inputs was required as total raw material usage for each product within the product range made over the production period was used. Products within the range were modelled individually for the declared unit of 1m². The Altro Classic 25 product obtains the highest results in all the results categories and it is these results which have been used on this EPD to represent the product range.

Secondary data has been drawn from the BRE LINA database v2.0.31 and the background LCI datasets are based on ecoinvent v3.2.

Cut-off criteria

No inputs or outputs have been excluded. All raw materials and packaging inputs, plus their transport, process and general energy and water use, production and non-production waste, have been included, except for direct emissions to air, water and soil, which are not measured.

LCA Results

Results per declared unit (1m²) of the 2.5mm thick Altro Classic 25 standard safety flooring without PUR lacquer, for the declared modules can be found in the following tables, and as the product which obtained the highest values in each result category, can be considered to represent the product range.

(MND = module not declared; MNR = module not relevant; INA = indicator not assessed; AGG = aggregated)

Parameters describing environmental impacts									
			GWP	ODP	AP	EP	POCP	ADPE	ADPF
			kg CO ₂ equiv.	kg CFC 11 equiv.	kg SO ₂ equiv.	kg (PO ₄) ³⁻ equiv.	kg C ₂ H ₄ equiv.	kg Sb equiv.	MJ, net calorific value.
Product stage	Raw material supply	A1	AGG	AGG	AGG	AGG	AGG	AGG	AGG
	Transport	A2	AGG	AGG	AGG	AGG	AGG	AGG	AGG
	Manufacturing	A3	AGG	AGG	AGG	AGG	AGG	AGG	AGG
	Total (of product stage)	A1-3	6.97	5.43e-7	0.0338	0.0117	7.16e-3	3.90e-5	149

GWP = Global Warming Potential;
 ODP = Ozone Depletion Potential;
 AP = Acidification Potential for Soil and Water;
 EP = Eutrophication Potential;

POCP = Formation potential of tropospheric Ozone;
 ADPE = Abiotic Depletion Potential – Elements;
 ADPF = Abiotic Depletion Potential – Fossil Fuels.

LCA Results (continued)

Parameters describing resource use, primary energy									
			PERE	PERM	PERT	PENRE	PENRM	PENRT	
			MJ	MJ	MJ	MJ	MJ	MJ	
Product stage	Raw material supply	A1	AGG	AGG	AGG	AGG	AGG	AGG	AGG
	Transport	A2	AGG	AGG	AGG	AGG	AGG	AGG	AGG
	Manufacturing	A3	AGG	AGG	AGG	AGG	AGG	AGG	AGG
	Total (of product stage)	A1-3	19.7	1.52e-4	19.7	174	0	174	

PERE = Use of renewable primary energy excluding renewable primary energy used as raw materials;
 PERM = Use of renewable primary energy resources used as raw materials;
 PERT = Total use of renewable primary energy resources;

PENRE = Use of non-renewable primary energy excluding non-renewable primary energy resources used as raw materials;
 PENRM = Use of non-renewable primary energy resources used as raw materials;
 PENRT = Total use of non-renewable primary energy resource

Parameters describing resource use, secondary materials and fuels, use of water

			SM	RSF	NRSF	FW
			kg	MJ net calorific value	MJ net calorific value	m ³
Product stage	Raw material supply	A1	AGG	AGG	AGG	AGG
	Transport	A2	AGG	AGG	AGG	AGG
	Manufacturing	A3	AGG	AGG	AGG	AGG
	Total (of product stage)	A1-3	0	0	0	0.478

SM = Use of secondary material;
RSF = Use of renewable secondary fuels;

NRSF = Use of non-renewable secondary fuels;
FW = Net use of fresh water.

Other environmental information describing waste categories

			HWD	NHWD	RWD
			kg	kg	kg
Product stage	Raw material supply	A1	AGG	AGG	AGG
	Transport	A2	AGG	AGG	AGG
	Manufacturing	A3	AGG	AGG	AGG
	Total (of product stage)	A1-3	0.324	0.527	2.93e-4

HWD = Hazardous waste disposed;
NHWD = Non-hazardous waste disposed;
RWD = Radioactive waste disposed.

LCA Results (continued)

Other environmental information describing output flows – at end of life

			CRU	MFR	MER	EE
			kg	kg	kg	MJ per energy carrier
Product stage	Raw material supply	A1	AGG	AGG	AGG	AGG
	Transport	A2	AGG	AGG	AGG	AGG
	Manufacturing	A3	AGG	AGG	AGG	AGG
	Total (of product stage)	A1-3	0	0.168	0.0945	0

CRU = Components for reuse;
MFR = Materials for recycling;

MER = Materials for energy recovery;
EE = Exported energy.

References

BSI. Sustainability of construction works – Environmental product declarations – Core rules for the product category of construction products. BS EN 15804:2012+A1:2013. London, BSI, 2013.

BSI. Environmental labels and declarations – Type III Environmental declarations – Principles and procedures. BS EN ISO 14025:2010 (identical to ISO 14025:2006). London, BSI, 2010.

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BSI. Resilient, laminate and textile floor coverings – Measurement of dynamic coefficient of friction on dry floor surfaces. BS EN 13893:2002. London, BSI, 2002.

DIN 51130: 2004 Testing of floor coverings; determination of slip resistance; work rooms and work areas subject to pronounced risk of slipping; walking method; ramp test German National Standard 2004.

CAN/ULC-S102.2, Standard Method of Test for Surface Burning Characteristics of Flooring, Floor Coverings and Miscellaneous Materials and Assemblies. Standards Council of Canada / Conseil canadien des normes.

ASTM E648, Test for Surface Burning Characteristics of Building Materials.